Joong-Gil Kim

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Research Interests

My research goal is to build robots that acquire complex control skills in simulation and carry them over to real hardware without losing reliability. I work on reinforcement learning for robotic manipulation and locomotion, high-fidelity physics simulation, and sim-to-real transfer. My research interests include deep reinforcement learning, dexterous manipulation, humanoid whole-body control, and scalable simulation frameworks for robot learning.

Education

Korea University of Technology and Education Mar 2025 – Feb 2027 (expected)
M.S. in Electronic Engineering (Advisor: Yong-Jae Kim) *Cheonan, Korea*

Korea University of Technology and Education Mar 2020 – Feb 2025
B.S. in Mechatronics Engineering *Cheonan, Korea*

Research Experience

IRIM Lab, Korea University of Technology and Education Sep 2024 – Present
Research Assistant (Advisor: Yong-Jae Kim) *Cheonan, Korea*

- Developing ALLEX, a humanoid upper-body robot: built its simulation framework and sim-to-real pipeline for learning-based control.

Neuromeka Jul 2024 – Aug 2024
Research Intern *Pohang, Korea*

- Built smart-factory simulation environments in NVIDIA Isaac Sim.

Control and Dynamic Systems Lab, Korea University of Technology and Education Sep 2023 – Aug 2024
Undergraduate Research Assistant (Advisor: Hong-keun Kim) *Cheonan, Korea*

Honors and Awards

Bronze Award (3rd Place), 2nd Koh Young AI Competition 2024
Held at ICCAS 2024, Jeju, Korea. For the 3-DOF inverted pendulum swing-up and stabilization project.

Grand Prize, UCLA RoMeLa Director's Award, Robot Utilization Idea Competition 2023
1st place out of 99 teams nationwide. For the Reusable Sky-Jump Robot.

3rd Place, Korea Association of Robot Industry Chairman's Award, R-BIZ Challenge 2023
ZEUS Industrial Robot ZERO Mission. For the Toppled Vial Cylinder Removal Robot.

Publications

[C2] **Joong-Gil Kim**, Wontae Ye, Geunwoo Cho, Seong-Ho Yun, Se-Hyoung Cho, and Yong-Jae Kim. *Comparative Study on Agility, Efficiency, and Impact Absorption of Bipedal Robots with Active Toes*, International Conference on Control, Automation and Systems (ICCAS), 2026.

[C1] Juhan Park, Taerim Yoon, Seungmin Kim, **Joong-Gil Kim**, Wontae Ye, Jeongeun Park, Yoonbyung Chai, Geonwoo Cho, Geunwoo Cho, Dohyeong Kim, Kyungjae Lee, Yong-Jae Kim, and Sungjoon Choi. *Learning Dexterous Grasping from Sparse Taxonomy Guidance*, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

Projects

ALLEX Simulation Model Release Jun 2026

Simulation lead, with WIRobotics

- Public release of the ALLEX simulation model, validated against the physical robot so that simulated and real behavior match down to contact forces.

3-DOF Inverted Pendulum Oct 2024

Team Pluto X, KOREATECH

- Swing-up and stabilization control of a three-degree-of-freedom inverted pendulum.

Reusable Sky-Jump Robot Dec 2023

Team project, KOREATECH

- Proposed a reusable design for the Sky-Jump jumping robot, extending the concept introduced by Prof. Dennis Hong (UCLA RoMeLa).

Toppled Vial Cylinder Removal Robot Oct 2023

Team project, KOREATECH

- Vision-guided robot that detects vials toppled over on a pharmaceutical vial-transfer conveyor line and removes them automatically, keeping the packaging process running without manual intervention. [video]

Technical Skills

Programming	Python (primary); MATLAB, C
Simulation	Isaac Sim, Isaac Lab, MuJoCo, Genesis, Newton
GPU Computing	NVIDIA Warp
Learning	PyTorch, reinforcement learning, imitation learning
Tools	Linux, Git, ROS 2, AWS